Counting Principles, Binomial Identities, and Mixed Problem Solving

N-20220720

§1.1 Adding cases and multiplying independent choices

The standard counting principles follow the same decomposition principles developed in MIT's *Mathematics for Computer Science*, especially the counting chapters on bijections and inclusion–exclusion.[1] We will treat each method as a rule for a different kind of combinatorial construction:

Addition principle. If a task can be done in disjoint cases with $m_1, m_2, \dots, m_n$ possibilities, then the total number is $m_1 + \cdots + m_n$.

Multiplication principle. If a task is done in stages with $m_1, m_2, \dots, m_n$ choices at each stage, then the total number is $m_1 m_2 \cdots m_n$.

Inclusion–exclusion. For sets $A_1, \dots, A_n$, one adds single sizes, subtracts pairwise intersections, adds triple intersections, and so on.

A Venn diagram in the note reviews the three-set formula:

$$\begin{aligned}|A \cup B \cup C| = {} & |A| + |B| + |C| \\ & - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.\end{aligned}$$

The addition principle applies to disjoint alternatives. The multiplication principle applies to successive independent choices. Many counting mistakes come from using multiplication when cases overlap, or using addition when a task has stages. A good habit is to write a sentence: "first choose ..., then choose ..." for multiplication, and "either ..., or ..." for addition.

The addition and multiplication principles are not merely classroom rules. They are the first operations on combinatorial classes. If $\mathcal{A}$ and $\mathcal{B}$ are disjoint classes, then

$$|\mathcal{A} \sqcup \mathcal{B}| = |\mathcal{A}| + |\mathcal{B}|.$$

If an object is built by choosing one object from $\mathcal{A}$ and one from $\mathcal{B}$, then

$$|\mathcal{A} \times \mathcal{B}| = |\mathcal{A}||\mathcal{B}|.$$

Thus addition corresponds to disjoint union, and multiplication corresponds to Cartesian product.

This viewpoint is useful because it explains why formulas should follow from constructions. Before computing a number, identify whether the object is a disjoint alternative, a staged product, a quotient by symmetry, or a coefficient in a generating function.

[1] MIT 6.042J, *Mathematics for Computer Science*, readings.

§1.2 Inclusion-exclusion in arithmetic counting

One problem asks how many positive integers not exceeding 2021 are neither multiples of 6 nor multiples of 8. The excluded set is

$$A \cup B,$$

where A is the set of multiples of 6, and B is the set of multiples of 8. Since

$$\operatorname{lcm}(6, 8) = 24,$$

we have

$$|A| = 336, \qquad |B| = 252, \qquad |A \cap B| = 84.$$

Thus the number excluded is

$$336 + 252 - 84 = 504,$$

and the answer is

$$2021 - 504 = 1517.$$

The calculation also has a short direct verification:

$$\#\{1 \le k \le 2021 : 6 \nmid k,\ 8 \nmid k\} = 2021 - \left\lfloor \frac{2021}{6} \right\rfloor - \left\lfloor \frac{2021}{8} \right\rfloor + \left\lfloor \frac{2021}{24} \right\rfloor = 1517.$$

Inclusion–exclusion corrects the double subtraction of common multiples.

The formula

$$|A \cup B| = |A| + |B| - |A \cap B|$$

is a small instance of Möbius inversion on the Boolean lattice. For a finite family $A_1, \dots, A_n$, the number of elements lying in at least one A_i is

$$\left| \bigcup_{i=1}^{n} A_i \right| = \sum_{\varnothing \neq S \subseteq [n]} (-1)^{|S|+1} \left| \bigcap_{i \in S} A_i \right|.$$

The alternating signs are the Möbius function of the subset poset:

$$\mu(S, T) = (-1)^{|T|-|S|}.$$

So inclusion–exclusion is not a trick for Venn diagrams; it is inversion of the operation that replaces exact membership data by cumulative membership data.

For a finite universe U, a set $A \subseteq U$ has indicator function

$$\mathbf{1}_A(x) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A. \end{cases}$$

Then

$$|A| = \sum_{x \in U} \mathbf{1}_A(x).$$

The two-set inclusion–exclusion identity is the pointwise identity

$$\mathbf{1}_{A \cup B} = \mathbf{1}_A + \mathbf{1}_B - \mathbf{1}_A \mathbf{1}_B.$$

Summing over U gives the cardinality formula.

This is also the beginning of probability. If X is uniformly chosen from U, then

$$\mathbb{P}(X \in A) = \frac{|A|}{|U|} = \mathbb{E}[\mathbf{1}_A(X)].$$

Counting and expectation become the same operation after normalization.

§1.3 Permutations, positions, and binomial coefficients

The page distinguishes arrangements and selections:

$$A_n^m = \frac{n!}{(n-m)!}, \qquad C_n^m = \frac{n!}{m!(n-m)!}.$$

A sequence problem asks for the number of 10-term sequences with $x + y = 10$ and $x - y = 4$, giving $x = 7, y = 3$. Therefore the number of sequences is

$$\binom{10}{3} = 120.$$

The thought process is: once the number of one kind of symbol is fixed, the sequence is determined by choosing its positions.

Counting problems involving subject choices, sign-chart inequalities, and similar-triangle configurations share one skill: case management. One must decide whether the structure is a sum of cases, a product of stages, or a coefficient extraction.

The binomial theorem is

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k.$$

Several standard identities follow immediately:

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n,$$

$$\sum_{k=0}^{n} (-1)^k \binom{n}{k} = 0,$$

and by differentiating

$$(1+x)^n$$

and setting $x = 1$,

$$\sum_{k=0}^{n} k \binom{n}{k} = n2^{n-1}.$$

The page writes these identities as examples of how a coefficient formula becomes a counting tool.

A worked example asks for the constant term or a particular coefficient in an expansion such as

$$(2x - 2y)^5.$$

The term involving $x^{5-k}y^k$ is

$$\binom{5}{k} (2x)^{5-k} (-2y)^k.$$

Thus the coefficient of x^2y^3 comes from $k = 3$:

$$\binom{5}{3} 2^2 (-2)^3 = -320.$$

The note's point is that coefficient problems become simple once the target exponent determines k.

The identity

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$$

comes from choosing a committee of size k from n people. Either a distinguished person is not chosen, or that person is chosen. This kind of proof is often clearer than algebraic manipulation because it explains what the two terms mean.

The binomial theorem says

$$(1+x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k.$$

The coefficient of x^k counts subsets of size k. Differentiating and setting $x = 1$ gives

$$\sum_{k=0}^{n} k\binom{n}{k} = n2^{n-1},$$

which counts pairs (S, s) where $S \subseteq [n]$ and $s \in S$: choose s first, then choose the rest of S.

For another concrete coefficient calculation, the coefficient of x^4 in $(1+x)^3(1+x^2)^2$ counts choices from three objects of weight one and two objects of weight two totaling weight four. It equals $\binom{3}{0}\binom{2}{2} + \binom{3}{2}\binom{2}{1} = 1 + 6 = 7$. Each factor records an independent choice and multiplication records the summed weight. This illustrates the right use of generating functions: algebraic operations on series encode operations on counted objects.

§1.4 Counting orbits under symmetry

If a finite group G acts on a finite set X, the number of orbits is

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |\operatorname{Fix}(g)|.$$

This is Burnside's lemma. It explains why one cannot always divide by the number of symmetries: objects with nontrivial stabilizers have smaller orbits.

For a square necklace of four binary beads, count colorings up to rotations. The identity fixes all $2^4 = 16$ colorings, the half-turn fixes $2^2 = 4$, and each quarter-turn fixes 2. Burnside's formula therefore gives

$$\#\text{binary length-four necklaces} = \frac{16 + 4 + 2 + 2}{4} = 6.$$

Simply dividing 16 by 4 would give the wrong answer because symmetric colorings have nontrivial stabilizers. This small calculation shows exactly when symmetry changes the unit of counting from individual objects to orbits.